CHAPTER 14

WAVES

CHAPTER ESSENCE AT A GLANCE:
Waves carry energy across space without transporting matter! When waves reflect in a pipe or string, forward and backward waves interfere to create stationary Standing Waves with still points (Nodes) and loud maximum-vibration points (Antinodes)!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Wave Types: Transverse vs Longitudinal:** Transverse: particle oscillation is perpendicular to wave propagation (light, waves on stretched string, crests & troughs). Longitudinal: particle oscillation is parallel to wave direction (sound in air, compressions & rarefactions).

• **Laplace Correction for Sound Speed:** Newton assumed isothermal sound propagation (v = sqrt(P divided by ρ) approx. 280 m/s, ~15% low). Laplace corrected that sound compressions are ADIABATIC (too fast for heat transfer) implies v = sqrt(γ P divided by ρ) = 331.2 m/s at STP (matches experiment!).

• **Standing Waves: Nodes & Antinodes:** Nodes (N) are points of ZERO displacement and maximum strain. Antinodes (A) are points of MAXIMUM displacement. Distance between adjacent Node and Antinode = λ divided by 4. Distance between adjacent Nodes = λ divided by 2.

• **Organ Pipe Harmonics:** Closed pipe (Node at closed end, Antinode at open end): produces ONLY ODD harmonics (1 : 3 : 5), ν n = (2n - 1) v divided by (4L). Open pipe (Antinodes at both ends): produces ALL harmonics (1 : 2 : 3 : 4), ν n = n v divided by (2L).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$y(x,t) = A\sin(kx - \omega t + \phi), \quad v = \frac{\omega}{k} = \nu\lambda$$

Harmonic Wave Function and Wave Velocity

$$v = \sqrt{\frac{T}{\mu}} \quad \text{and} \quad v = \sqrt{\frac{\gamma P}{\rho}}$$

Speed of Transverse String Wave and Newton-Laplace Sound

$$\nu_n = n\frac{v}{2L} \quad (n = 1, 2, 3 \dots) \quad (\text{Stretched String / Open Pipe})$$

Harmonics of String and Open Organ Pipe

$$\nu_n = (2n-1)\frac{v}{4L} \quad (n = 1, 2, 3 \dots) \quad (\text{Closed Organ Pipe})$$

Odd Harmonics in Closed Organ Pipe

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Closed pipes produce ONLY odd harmonics (1st, 3rd, 5th). Open pipes produce ALL harmonics (1st, 2nd, 3rd, 4th), giving open pipes a richer musical tone!

• **Alert:** At a closed end of a pipe, air cannot move so it forms a NODE (N). At an open end, air moves freely forming an ANTINODE (A)!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Differentiate between transverse waves and longitudinal waves with one example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Transverse Waves]** Medium particles oscillate perpendicular to the direction of wave propagation. Can propagate only in solids and liquid surfaces (e.g. Light waves, vibrations in a plucked guitar string). [1 Mark]

• **[Step 2: Longitudinal Waves]** Medium particles oscillate parallel to the direction of wave propagation through compressions and rarefactions. Can propagate through solids, liquids, and gases (e.g. Sound waves in air). [1 Mark]

• **Key Exam Result:** Transverse (perpendicular oscillation) vs Longitudinal (parallel oscillation)

2. Write Newton's formula for the speed of sound in a gas. Why did Laplace correct it? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Newton's Formula & Flaw]** Newton assumed sound propagation is an isothermal process (B = P): v = √(P divided by ρ). At STP in air, this gives v = √(1.013 × 10^5 divided by 1.29) ≈ 280 m/s, which is 15% lower than the experimental value (332 m/s). [1 Mark]

• **[Step 2: Laplace Correction]** Laplace pointed out that compressions and rarefactions occur so rapidly that heat cannot be exchanged with surroundings, making the process ADIABATIC (B = γ P). Corrected formula: v = √(γ P divided by ρ). At STP, this yields v ≈ 331.3 m/s, matching experiments perfectly. [1 Mark]

• **Key Exam Result:** Newton: v = sqrt(P divided by ρ) (Isothermal); Laplace: v = sqrt(γ P divided by ρ) (Adiabatic)

3. What are beats? Explain how beats are produced and write the expression for beat frequency: nu beat = nu1 - nu2. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Phenomenon of Beats]** Beats is the periodic waxing and waning (variation in loudness) of sound heard when two sound waves of slightly different frequencies travel in the same direction and superpose. [1 Mark]

• **[Step 2: Analytical Derivation]** Let y1 = A sin(2 π nu1 t) and y2 = A sin(2 π nu2 t). By superposition: y = y1 + y2 = 2 A cos[2 π (nu1 - nu2) t divided by 2] sin[2 π (nu1 + nu2) t divided by 2]. The resultant amplitude is A mod = 2 A cos[π (nu1 - nu2) t]. [1 Mark]

• **[Step 3: Beat Frequency]** Intensity is maximum when |cos[π (nu1 - nu2) t]| = 1, occurring at intervals Δt = 1 divided by (nu1 - nu2). The number of beats per second (beat frequency) is: nu beat = 1 divided by Δt = nu1 - nu2. [1 Mark]

• **Key Exam Result:** Beat Frequency nu beat = |nu1 - nu2|

4. Derive expressions for the normal modes of vibration and frequencies in a stretched string fixed at both ends. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Boundary Conditions]** For a stretched string of length L fixed at x = 0 and x = L, nodes are formed at both ends: y(0, t) = 0 and y(L, t) = 0. Standing wave: y(x, t) = 2 A sin(k x) cos(ω t). [1 Mark]

• **[Step 2: Allowed Wavelengths]** At x = L: sin(k L) = 0 => k L = n π => (2 π divided by lambda n) L = n π => lambda n = 2 L divided by n (where n = 1, 2, 3...). [1 Mark]

• **[Step 3: Allowed Frequencies]** Frequency nu n = v divided by lambda n = n (v divided by 2L) = n nu 1, where v = sqrt(T divided by μ). For n = 1: Fundamental frequency nu 1 = v divided by 2L. Frequencies are in integer ratio: nu1 : nu2 : nu3 = 1 : 2 : 3 (all harmonics present). [1 Mark]

• **Key Exam Result:** nu n = n (v divided by 2L); Harmonics ratio = 1 : 2 : 3 : ...

5. Derive expressions for the harmonic frequencies of standing waves in an Open Organ Pipe of length L. Why is an open pipe musically richer than a closed pipe? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Boundary Conditions in Open Pipe]** In an organ pipe open at both ends, antinodes are formed at both open ends. Distance between consecutive antinodes is λ divided by 2. L = n (lambda n divided by 2) => lambda n = 2 L divided by n (for n = 1, 2, 3...). [1.5 Marks]

• **[Step 2: Frequencies & Musical Richness]** Frequencies nu n = v divided by lambda n = n (v divided by 2L) = n nu 1. All harmonics (1st, 2nd, 3rd, 4th...) are generated in ratio 1 : 2 : 3 : 4... In contrast, a closed organ pipe generates ONLY odd harmonics (1 : 3 : 5...). Because open pipes produce both even and odd harmonics, they have a richer, more melodic timbre. [1.5 Marks]

• **Key Exam Result:** nu n = n (v divided by 2L); Contains all harmonics (1:2:3:4), hence richer tone

6. Two tuning forks A and B produce 5 beats per second. When fork A is filed slightly, the beat frequency increases to 7 beats/s. If the frequency of fork B is 256 Hz, find the original frequency of fork A. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Possible Frequencies for A]** Beat frequency = |nu A - nu B| = 5 Hz. Given nu B = 256 Hz: nu A = 256 + 5 = 261 Hz OR nu A = 256 - 5 = 251 Hz. [1 Mark]

• **[Step 2: Effect of Filing]** When a tuning fork is filed, its mass decreases, which INCREASES its frequency (nu A increases). [1 Mark]

• **[Step 3: Deduce Correct Frequency]** If nu A was 261 Hz: as nu A increases (e.g. to 263 Hz), new beat frequency = 263 - 256 = 7 Hz (INCREASES, matching given data). If nu A was 251 Hz: as nu A increases (e.g. to 253 Hz), new beat frequency = 256 - 253 = 3 Hz (DECREASES, contradictory). Therefore, original frequency of A must be 261 Hz. [1 Mark]

• **Key Exam Result:** Original Frequency of Fork A = 261 Hz

7. A pipe 20 cm long is closed at one end. Which harmonic mode of the pipe is resonantly excited by a 425 Hz source? (Speed of sound in air = 340 m/s) [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Fundamental Frequency of Closed Pipe]** Length L = 20 cm = 0.20 m, speed of sound v = 340 m/s. Fundamental frequency ν 1 = v divided by (4 L) = 340 divided by (4 × 0.20) = 340 divided by 0.80 = 425 Hz. [1.5 Marks]

• **[Step 2: Harmonic Mode Identification]** Since the exciting source frequency is 425 Hz, which equals the fundamental frequency nu 1, the pipe is resonating in its FUNDAMENTAL MODE (First Harmonic). [1.5 Marks]

• **Key Exam Result:** Resonantly excited in First Harmonic (Fundamental Mode, n = 1)